Changes in PTM abundance indicate temporal damage accumulation in the surrounding regions of PLF identified protein loci exhibiting significant age-associated structural differences

Methods:

Total protein PTMs identified across each biological replicate during the regeneration of the protein abundance data via Mascot was recorded. PTMs specific to regions of significant age associated PSC difference or nearby regions of interest like endogenous cleavage sites were mapped to their corresponding 50aa peptide region through the construction of a PTM localisation tool. This tool was created using the previously stated development environment. Mascot .csv search results are read into the tool and a manually set aa window is specified by the user to identify the abundance of all or specific PTMs. This built for purpose tool currently only searches for Oxidation (K, P, M) deamidation (NQ), galactosyl and glucosylgalactosyl glycosylations of K. Total PTM counts were summed and averaged (mean) across the 3 specimen samples per treatment group (eg. 3x fuzzy female rats) across all peptides associated to that region. This was performed for the regions of significant PSC difference and the surrounding region +- 50aa to account for regions of interest bordering the significant 50aa region. Mann-Whitney U test for significance was used to assess statistical differences across these independent groups. These results were visualised in bar charts and all detected modifications were mapped across the studied region, the supplementary results of which are stored in Github repository...

Results and Discussion

Significant changes in average PTM abundance across all peptides mapped to PLF identified protein 50aa loci exhibiting significant age-associated structural differences were not seen throughout the study. Observation of PTM abundance in the surrounding adjacent 50aa regions were also not seen considered to be statistically significant by Mann-Whitney U non-parametric test for significance. However, in the surrounding regions 50aa regions of significant age associated difference mentioned throughout this study, seemingly large changes in average PTM abundance was observed. Upon further analysis, this time assuming non-independence between sample groups statistical significance was observed in a variety of proteins mentioned in (S2 and S3). Further rudimentary insights were explored but lie beyond the scope of this study.

Analysing the total and average PTM count in the aa250-200 +-50aa region an average increase in Oxidation (K, P, M) and Deamidation (NQ) was observed (S3) in Jumbo rats compared to fuzzy specimens. These increases were not statistically significant but notable. It is important to note that very few PTMs were detected within the PLF identified 50aa regions of interest but in the surrounding +- 50aa regions. This trend continues throughout the PTM analysis. The MPSC tool however did not indicate these regions as hugely susceptible to ROS. The average PTM counts found in the aa950-100 +-50aa region were very highly similar in both jumbo and fuzzy specimens and likewise in the aa1050-1100 +-50aa region.

The observation of average PTM counts at aa1000-1050 +-50aa and 1050-1100 +-50aa shows signs of increased oxidative stress. The total average PTMs count increased for oxidation (K, P) and deamidation in jumbo rats, however no observations of oxidation (M) were present in these regions. Again, the vast majority of PTMs were localised to the surrounding 50aa regions of the PLF identified region of significant difference. The increase in deamidation again signals a reduction in structural integrity seen in CO1A2 reinforcing the idea of increased T1Coll degradation in jumbo rats, interestingly CO1A2 peptides were devoid of oxidation (M) in this study, which is slightly unexpected considering the increased MPSC predicted susceptibility of human CO1A2, displayed in Figure 5B.

In CO2A1 very low numbers of PTMs were identified. Generally higher average PTM counts were observed in fuzzy specimens but they were not very different to the Average PTMs identified in jumbo specimens.

Limitations:

Statistical Independence – The Mann-Whitney U test was used to assess statistically significant differences between the average total PTM counts between fuzzy and jumbo specimen groups. However, further analysis utilising the Wilcoxon test, assuming dependency at first glance appears to provide more insightful representation of the data (see S3).